

Adiabatic demagnetization as a pathway toward cryogenic and ambient magnetic refrigeration

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Abstract

With global temperatures steadily increasing, conventional cooling technologies, which consume substantial energy and use damaging refrigerants, now significantly worsen the warming they are meant to reduce. Magnetic refrigeration which operate on the principle of magnetocaloric effect (MCE) and adiabatic demagnetization refrigeration (ADR), offers an efficient and environmentally friendly alternative. This project investigates the low-temperature magnetocaloric behaviour of $\text{Gd}_3\text{Te}_2\text{Li}_3\text{O}_{12}$ (LGTO), a geometrically frustrated Gd^{3+} -based garnet with weak spin correlations and a suppressed ordering temperature, making it a strong ADR candidate. A temperature as low as 165 mK with a holding time of more than 12 hours is achieved making it an alternate to current He-3 based cooling. In addition, the study explores tunable intermetallic systems capable of forming high-entropy alloys (HEAs) with room-temperature magnetic transitions, highlighting their potential as sustainable refrigerants for ambient-temperature applications.

1. Introduction

Cryogenic and room temperature magnetic refrigeration technologies are gaining attention as sustainable alternatives to conventional gas based cooling systems, which rely on CFC or HFC type refrigerants that contribute to global warming and environmental degradation. Magnetic refrigeration, instead, exploits the magnetocaloric effect, where a material's temperature changes in response to an applied magnetic field. The present work explores two promising material classes for such applications: (i) geometrically frustrated Gd^{3+} based garnets suitable for adiabatic demagnetization at ultra low temperatures, and (ii) tunable high entropy intermetallic alloys capable of exhibiting giant magnetocaloric effects near room temperature. Together, these projects focus on finding alternate materials for environment friendly refrigeration.

1.1. Frustrated Gd^{3+} garnets as candidates for adiabatic demagnetization refrigeration: The case of $\text{Gd}_3\text{Te}_2\text{Li}_3\text{O}_{12}$

Ultra low temperatures are essential for advancements in diverse fields such as quantum computing, condensed matter physics etc. Such temperatures have been traditionally attained using liquid helium-based cooling technology. However, the global shortage of helium and the escalating cost of its He^{3+} isotope have created an emerging need for alternative cryogenic technologies [1]. Among the most promising substitutes are water-free solid-state magnetic refrigeration systems which operate based on the principle of adiabatic demagnetization refrigeration (ADR)/magnetocaloric effect (MCE). MCE refers to the reversible temperature change in a magnetic material under an applied magnetic field which arises due to the entropy change associated with spin alignment. For efficient magnetic refrigeration at sub-Kelvin temperatures, materials must possess a large magnetic entropy change ($-\Delta S_M$), low magnetic ordering temperatures, and weak spin correlations. In this regard, Gd^{3+} -based compounds are particularly attractive. The Gd^{3+} ion has a half-filled 4f shell with a high-spin ground state ($S = 7/2$) and negligible orbital contribution ($L = 0$), resulting in nearly isotropic Heisenberg-type exchange interactions and minimal magnetic anisotropy. These characteristics favor strong magnetocaloric performance and reversible entropy changes across wide temperature ranges.

The candidate material gadolinium lithium tellurate garnet, $\text{Gd}_3\text{Te}_2\text{Li}_3\text{O}_{12}$ (LGTO), crystallizes in the cubic $Ia\bar{3}d$ space group and exhibits a three-dimensional network of corner-sharing Gd^{3+} triangles, widely known as hyper-Kagome lattice. Such a geometry gives rise to geometrical frustration, wherein all pairwise exchange interactions

cannot be simultaneously satisfied. This frustration suppresses long-range magnetic ordering, leading to a highly degenerate magnetic ground state and significant magnetic entropy retention down to low temperatures. Consequently, this frustrated Gd^{3+} garnet represent promising candidates for adiabatic demagnetization refrigeration (ADR) near the liquid helium regime.

1.2 High entropy alloys and giant magnetocaloric materials

With conventional refrigeration systems contributing significantly to environmental pollution through the emission of chlorofluorocarbons (CFCs), and the need for silent, vibration-free cooling in applications such as submarines and covert operations, the search for alternative refrigeration technologies has gained importance. Among these, magnetic refrigeration has emerged as a promising future replacement owing to its energy efficiency, environmental friendliness, and operational silence. A particularly interesting class of such materials are those exhibiting the giant magnetocaloric effect (GMCE), characterized by a large field-induced isothermal entropy change ($-\Delta S_M$) and adiabatic temperature variation (ΔT_{ad}). To achieve room-temperature magnetocaloric performance, materials must possess two structural phases with substantially different magnetic properties: Curie temperature (T_C) and structural transition temperature (T_{st}) which can be tuned to lie close to ambient temperature. The magnitude of GMCE is strongly associated with the presence of magnetostructural transitions, where magnetic and structural phase transitions coincide, leading to large entropy changes.

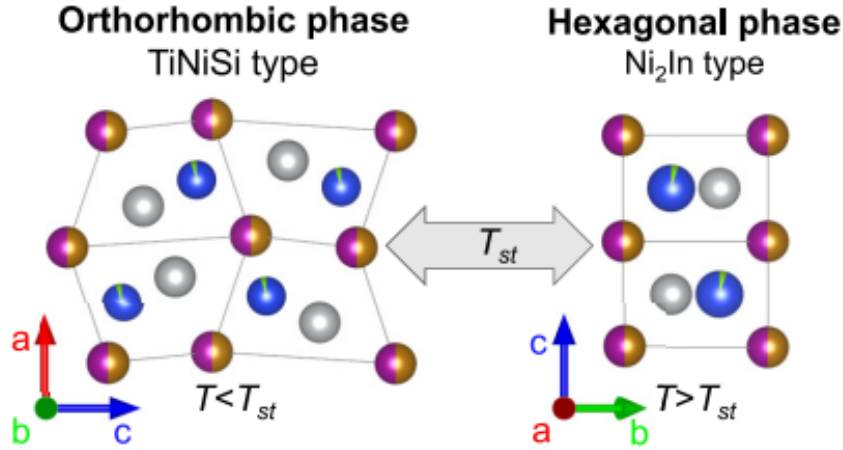


Figure 1: *Magnetostructural transitions in prototype $MM'X$ materials [2].*

High-entropy alloys (HEAs) are particularly attractive in this context because their large configurational entropy stabilizes simple solid-solution phases, allowing wide chemical tunability of the stoichiometry of multiple components. This tunability enables precise control over magnetic exchange interactions, structural transitions, and electronic environments, all of which are crucial for engineering room-temperature magnetocaloric performances. Their high lattice and magnetic entropy also opens the possibility of achieving a GMCE when the composition is suitably tuned to couple magnetic and structural instabilities. Moreover, HEAs often exhibit enhanced mechanical stability, corrosion resistance, and thermal robustness, making them suitable for repeated magnetic cycling in refrigeration applications.

In multicomponent systems, the formation of a high entropy alloy competes directly with the formation of ordered intermetallic compounds. Whether the system stabilizes as a single-phase HEA or decomposes into intermetallic phases depends critically on parameters such as the atomic size mismatch (δ), enthalpy of mixing (ΔH_{mix}), and configurational entropy (ΔS_{mix}). For HEA formation, these quantities must fall within specific threshold windows; otherwise, the alloy may partially or fully segregate into intermetallic phases. Thus, ensuring that the calculated values lie within a particular threshold limits, is essential for obtaining a stable single-phase HEA rather than a mixture of intermetallics.

Among potential candidates, Mn-based intermetallics (MnTX , where $T = \text{Ni, Fe, Co}$ and $X = \text{Si, Ge, Sn}$) exhibit strong magnetostructural coupling [2]. In the MnNiSi system ($T_{st} \approx 1206 \text{ K}$, $T_C \approx 622 \text{ K}$), chemical substitutions can tune both transitions toward room temperature, enhancing the magnetocaloric response. For instance,

in $\text{Fe}_{0.5}\text{Mn}_{0.5}\text{NiSi}_{1-x}\text{Al}_x$, varying the Al content aligns T_C and T_{st} near ambient conditions. The coexistence of orthorhombic and hexagonal phases in these alloys yields complex exchange interactions, where each phase contributes differently to the magnetic transition. This phase interplay enables fine control over magnetocaloric behavior and supports the realization of near room-temperature giant MCE.

The goal of the second part of the work is to identify and tune MnTX materials to form HEAs that exhibit room-temperature magnetocaloric effects using both first-principles (DFT) calculations and experimental validation.

2. Methodology

2.1 Experimental details for $\text{Gd}_3\text{Te}_2\text{Li}_3\text{O}_{12}$

Polycrystalline samples of Gadolinium Lithium Tellurate Garnet ($\text{Gd}_3\text{Te}_2\text{Li}_3\text{O}_{12}$, LGTO) were synthesized by the conventional solid-state reaction method. High-purity compounds of lithium carbonate (Li_2CO_3), tellurium dioxide (TeO_2), and gadolinium oxide (Gd_2O_3) were used as starting materials. Stoichiometric amounts of the reactants were weighed according to the nominal composition. The weighed powders were thoroughly mixed and homogenized using an agate mortar and pestle in acetone medium for several hours to ensure uniformity. The resulting mixture was pelletized under uniaxial pressure and subjected to initial sintering in air at 773 K for 5 hours. Afterwards, the mixture was reground for 2 hours, made into pellets and kept for the second heat treatment at 1173 K for 9 hours. To achieve phase homogeneity, the sintered material was repeatedly ground, pelletized, and re-sintered under identical thermal conditions for 2-3 cycles.

Phase identification of the synthesized LGTO was carried out by powder X-ray diffraction (XRD) using a *PAN Analytical X-Ray* diffractometer equipped with $\text{Cu-K}\alpha$ radiation. The diffraction data were collected in the 2θ range of 10° - 90° at room temperature. Rietveld refinement of the XRD data was performed using the *FULLPROF* software package to confirm the formation of the cubic garnet phase and to extract lattice parameters. Magnetic measurements were performed using a Quantum Design Physical Property Measurement System (PPMS) equipped with a Vibrating Sample Magnetometer (VSM). Temperature-dependent magnetization $M(T)$ and isothermal magnetization $M(H)$ measurements were recorded in the temperature range of 2-300 K under various applied magnetic fields. Additionally, adiabatic demagnetization refrigeration (ADR) measurements were performed using the custom built home-made setup to evaluate the magnetocaloric performance of LGTO. Time, temperature, and magnetic field variations during ADR cycles were recorded and analyzed to estimate the magnetic Grüneisen parameter to identify possible field-induced transitions.

2.2 Calculations for a suitable high entropy alloy

Different critical parameters were calculated to get nominal composition compounds in MnTX family. The calculated parameters yielded $\text{Fe}_{0.5}\text{Mn}_{0.5}\text{NiSi}_{0.95}\text{Ge}_{0.05}$ as a good candidate. The configurational entropy of mixing (ΔS_{mix}) for the alloy $\text{Fe}_{0.5}\text{Mn}_{0.5}\text{NiSi}_{0.95}\text{Ge}_{0.05}$ was calculated using the standard relation for ideal random mixing:

$$\Delta S_{\text{mix}} = -R \sum_i c_i \ln c_i$$

where R is the universal gas constant and c_i represents the atomic fraction of each constituent element. The obtained value of configurational entropy ($\Delta S_{\text{mix}} = 1.39R$) is slightly lower than the empirical criterion of $1.5R$ generally considered necessary for the stabilization of high-entropy alloys [4].

The enthalpy of mixing value calculated using the expression:

$$\Delta H_{\text{mix}} = \sum_{\substack{i=1 \\ i \neq j}}^N \Omega_{ij} c_i c_j$$

where Ω_{ij} represents the interaction parameter between elements i and j , and c_i, c_j are their respective atomic frac-

tions. The obtained value of ΔH_{mix} (9.301045 J/mol) slightly exceeds the upper limit for high-entropy alloy formation (7.5 J/mol) as reported in [3]. The atomic radius mismatch (δ) was calculated using the standard relation:

$$\delta = \sqrt{\sum_{i=1}^N c_i \left(1 - \frac{r_i}{\bar{r}}\right)^2} \times 100\%, \quad \bar{r} = \sum_{i=1}^N c_i r_i,$$

The obtained value of δ ($\approx 6.01\%$) was found to be less than the empirical threshold required for the formation of a single-phase high-entropy alloy, as reported in [3]. This indicates that the present composition possesses an acceptable degree of lattice distortion, favoring the stability of a solid-solution phase.

3. Results and Discussions

The room-temperature powder XRD pattern of LGTO and the final refinement parameters obtained using the FULLProf software are shown in Fig. 2. The diffraction profile exhibits no impurity peaks, and all reflections match well with the standard pattern, confirming that LGTO is formed in a single phase with a cubic $Ia\bar{3}d$ space group.

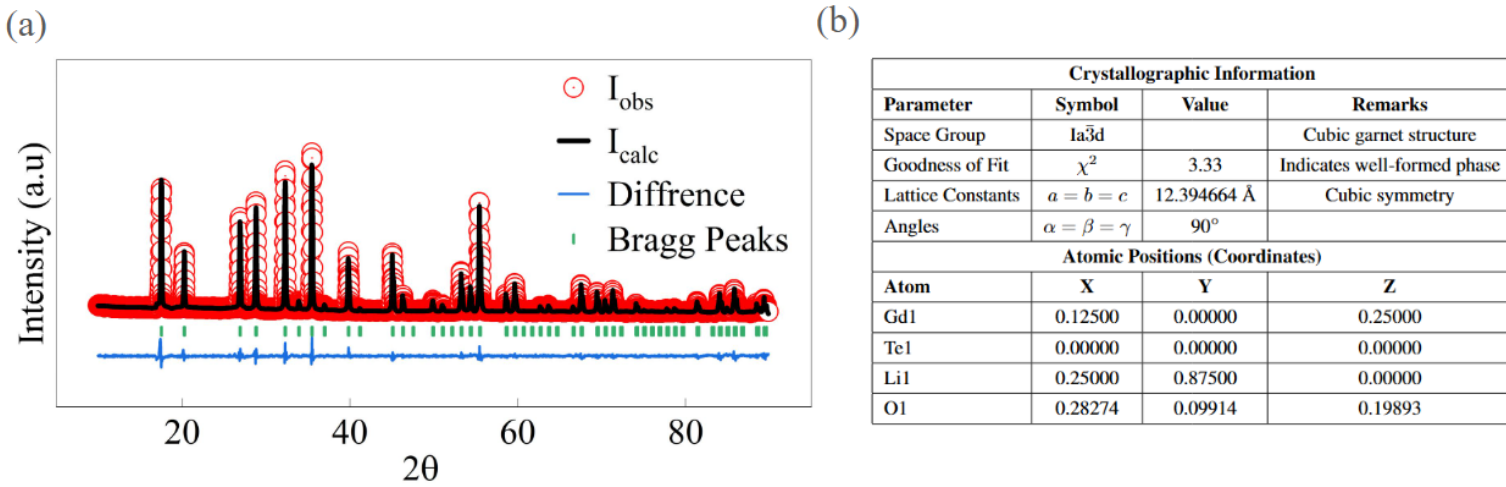


Figure 2: (a) Room-temperature XRD pattern of LGTO along with the Rietveld refinement obtained using the FULLProf package. Red open circles and black solid curve show the experimental and calculated data respectively. The vertical green lines and blue solid lines depict Bragg planes and difference in experimental and theoretical data, respectively. (b) Final structural refinement parameters corresponding to the best fit.

The CIF file obtained after phase identification was visualized using the VESTA software to understand the crystal structure, as shown in Fig.3. From the figure, it can be inferred that LGTO contains a high density of magnetic sublattices, as nearly every cationic site is occupied by Gd^{3+} ions. The Gd ions form a three-dimensional hyper-Kagome lattice consisting of a network of corner sharing triangles (Fig. 3 (a)). Owing to these corner-sharing triangular units, the antiferromagnetic exchange interactions cannot be simultaneously satisfied, preventing the spins from settling into a unique ordered configuration. This geometric frustration can possibly suppress the long-range magnetic order and, leads instead, to weak spin correlations over a wide temperature range. In addition to geometric frustration, the large Gd-Gd separations and the presence of nonmagnetic Te^{6+} and Li^+ linkers result in weak super-exchange pathways. Consequently, the effective exchange coupling constant (J) will be small, such that the exchange energy scale is easily dominated by thermal fluctuations ($k_B T$) at moderate temperatures. This can further suppress magnetic ordering and push the ordering temperature to very low values.

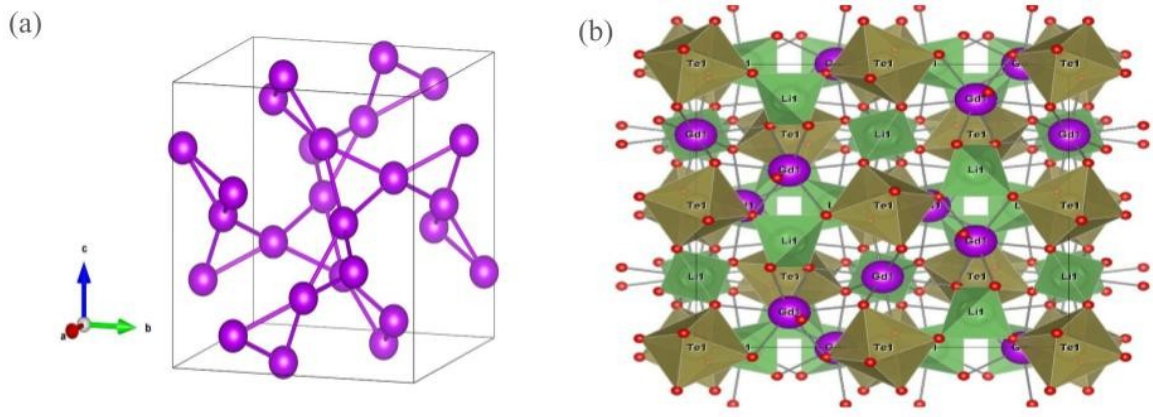


Figure 3: a) Magnetic Gd^{3+} ions located at the vertices of corner-sharing triangles. b) Crystal structure of LGTO.

To characterize the magnetic ground state and verify the implications of the frustrated exchange network described above, temperature-dependent magnetization measurements were performed under both zero-field-cooled (ZFC) and field-cooled (FC) protocols down to 1.8 K. The left panel of Fig.4 (a) shows the ZFC and FC susceptibility curves, which clearly indicate the absence of peak corresponding to magnetic ordering or spin-glass-like behaviour down to 1.8 K as there is no bifurcation between ZFC and FC curves. To quantify the strength and nature of the magnetic interactions, the inverse susceptibility data was analysed using the Curie-Weiss law. The Curie-Weiss fit for the inverse susceptibility data is shown in the right panel of Fig 4 (a). The fit yields $\theta_{CW} = -2.50$ K and $C = 24.3 \text{ emuK mole}^{-1}$. The effective magnetic moment, calculated using $\mu_{\text{eff}}(\mu_B) = 2.828 \sqrt{C}$, gives $\mu_{\text{eff}} = 8.0339 \mu_B$. The theoretical value, obtained from $\mu_{\text{eff}} = g\sqrt{J(J+1)}$, is $\mu_{\text{eff}} = 7.9373 \mu_B$, showing good agreement with the experimental result.

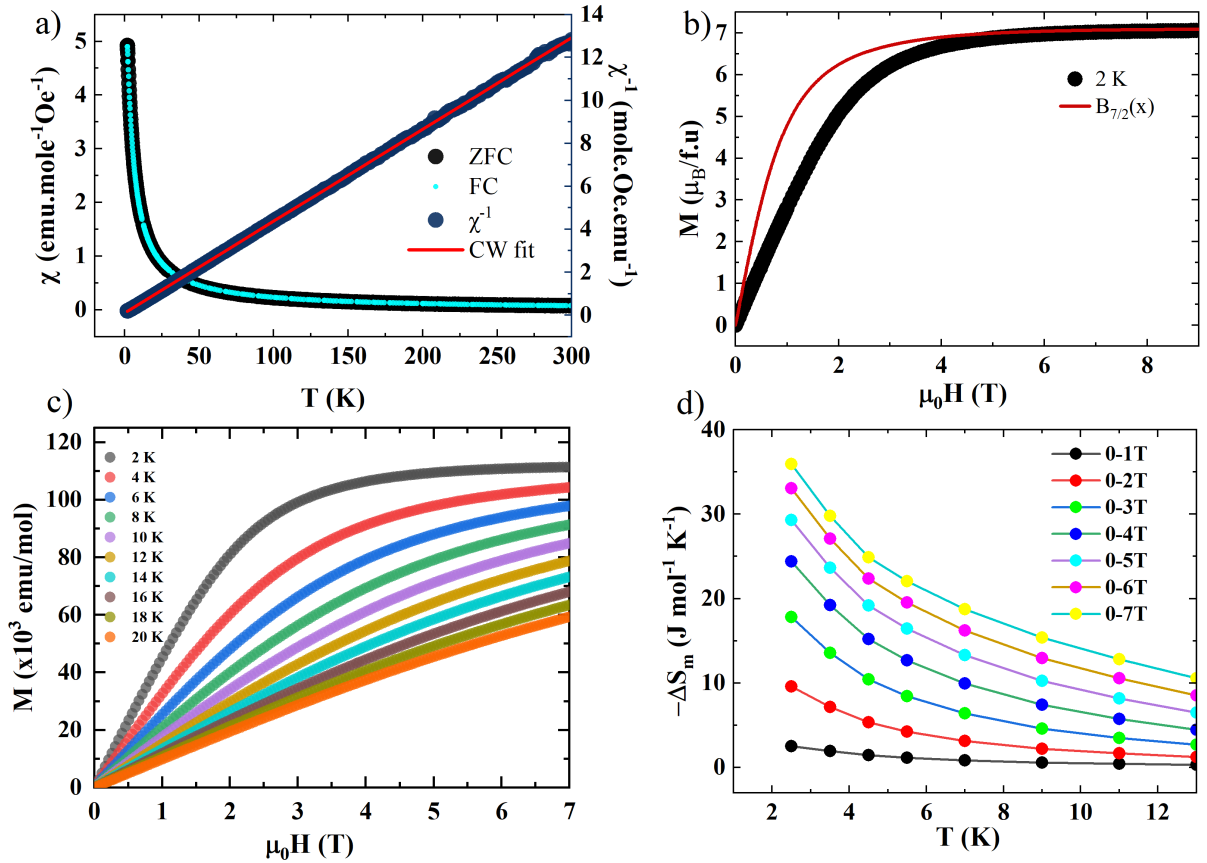


Figure 4: (a) Temperature variation of susceptibility (χ) and inverse susceptibility (χ^{-1}) with Curie-Weiss fitting (red line). (b) Isothermal magnetization versus applied field $M(H)$ at 2 K with Brillouin fitting, shown by solid red curve. (c) Isothermal magnetization vs. applied field $M(H)$ curves at various temperatures (2–20 K). (d) Magnetic entropy change ($-\Delta S_M$) for LGTO as a function of temperature in the range 2–12 K

Additionally, the negative sign of θ_{CW} signifies the presence of antiferromagnetic interactions in LGTO. Further, as no ordering is observed around θ_{CW} , it indicates the presence of magnetic frustration arising due to hyper-Kagome

geometry.

Further, the exchange interaction parameter calculated using the following equation:

$$J_{ex} = \frac{3k_B\theta_{CW}}{zS(S+1)} = -0.059K,$$

is negative, suggesting weak antiferromagnetic interactions among the magnetic ions. Afterwards, magnetic field dependence of the magnetization was measured in order to probe the possibility of any kind of field induced behavior. The M - H isotherms reveal how the spins respond to an applied magnetic field when thermal fluctuations are minimal. Therefore, M - H was measured at 2 K, a temperature (less than $\theta_{CW} = -2.5K$) low enough to suppress thermal disorder, as shown in Fig. 4 (b). The 2 K isotherm was fitted using the Brillouin function, which considers non-interacting spins corresponding to a paramagnetic system. The fit matches well in the high-field region but shows a noticeable deviation at low fields. Such deviations are typically associated with the development of antiferromagnetic correlations. The magnetization starts saturating at a field around 6 T. To further investigate the field and temperature dependence of the possible magnetic correlations, M - H isotherms at different temperatures are shown in Fig. 4 (c). It was found that as the temperature increases, the curves gradually approach Brillouin-like behaviour, consistent with dominant paramagnetic contributions.

For a quantitative understanding of the magnetocaloric behaviour, the magnetic entropy change ΔS_m was obtained using the relation:

$$\Delta S_m(T, H) = \int_0^H \left(\frac{\partial M}{\partial T} \right)_H dH.$$

The derivative $(\partial M/\partial T)_H$ was evaluated from the M - H isotherms measured in the first quadrant, and the magnetic entropy change $-\Delta S_m$ was calculated by integrating for different field intervals. The magnitude and temperature dependence of $-\Delta S_m$ provide insight into how efficiently the spins respond to an applied magnetic field. For low and intermediate fields, $-\Delta S_m$ decreases smoothly with temperature, consistent with predominantly paramagnetic behaviour governed by thermal disorder and weak antiferromagnetic correlations. However, at higher fields the $-\Delta S_m(T)$ curves begin to develop a more pronounced peak-like behaviour, with the magnitude increasing rapidly. Such an evolution of the entropy profile indicate the field induced polarization of spins, where the applied magnetic field drives the system closer to a crossover or instability in the magnetic energy landscape. The non-saturating large value of $-\Delta S_m$ at the lowest temperature for 7 T field signifies large entropy accumulation at temperatures lower than 1.8 K, suggesting the possibility of achieving very low temperatures in the ADR test.

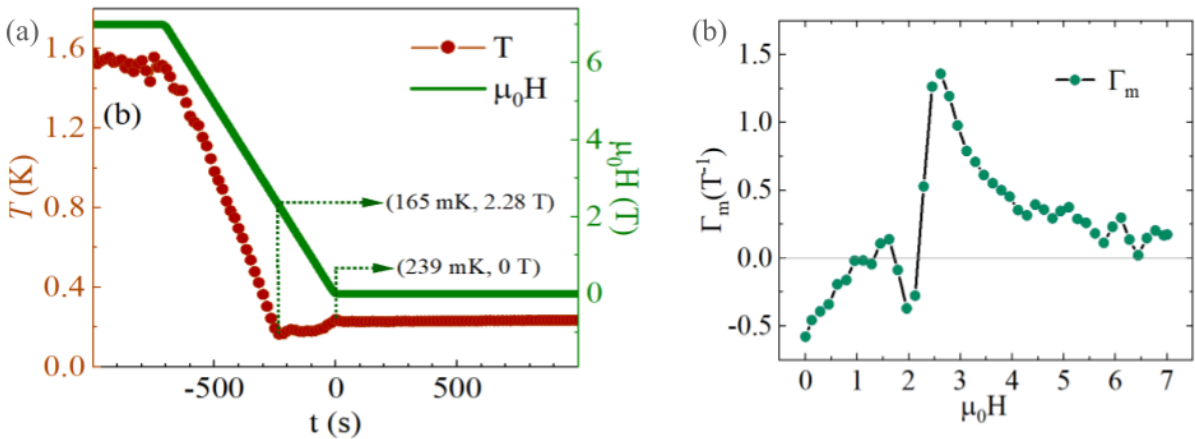


Figure 5: a) Left panel: sample temperature as a function of time; right panel: magnetic field variation with time b) Grüneisen parameter as a function of magnetic field

To directly measure the temperature change and holding time of LGTO, ADR measurements were performed by monitoring the variation of temperature and magnetic field as a function of time during a demagnetization cycle. The ADR data is shown in Fig.5 which shows the time evolution of sample temperature as a function of time (in the left panel) when the magnetic field is adiabatically reduced (right panel). It reveals a minimum attainable temperature of approximately 165 mK at an applied field of 2.28 T and a holding time exceeding 12 hours. As the field reaches zero,

the sample temperature slightly increases to 239 mK. This behaviour is consistent with a highly frustrated magnetic system, where the suppression of long-range magnetic order to very low temperatures enables efficient adiabatic demagnetization. Notably, the fact that the lowest temperature is achieved at a finite field, rather than strictly at zero field, suggests that the system undergoes a field induced changes in the magnetic state. This further suggests the enhancement of entropy at the finite field of 2.28 T. In order to understand such behavior, magnetic Grüneisen parameter has proven to be a vital quantity. It measures the sensitivity of entropy to an external control parameter; in this case, the applied magnetic field. For a field-driven process, it is defined as:

$$\Gamma_{\text{mag}} = \frac{1}{T} \left(\frac{\partial T}{\partial H} \right)_S$$

which relates the change in temperature with field under adiabatic conditions. Divergence-like behaviour or a sign change in Γ_{mag} is an indicator of field-induced transitions or proximity to a critical point. Fig. 5 (b) depicts the field dependence of Γ_{mag} evaluated from the ADR data and the observed sign change confirms the presence of field-induced transition.

4. Future direction

Future work will focus on a more detailed mapping of these field-induced transitions, locating the associated critical points, and probing the system at even lower temperatures to explore the possible existence of a quantum critical end point. In parallel, first-principles DFT calculations will be carried out to estimate the Curie temperature (T_C) of candidate high-entropy alloys, followed by the synthesis and characterization of different stoichiometries to identify materials capable of exhibiting room-temperature magnetocaloric performance for practical magnetic refrigeration applications.

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